

Operator	Graphcore	CS-3	Groq
aten.abs	C	C	C
aten.absolute	C	C	C
aten.add.Scalar	C	C	C
aten.add.Tensor	C	C	C
aten.addcddiv	C	C	R
aten.addcmul	C	C	C
aten.bitwise_and.Scalar	F	F	F
aten.bitwise_and.Tensor	F	F	F
aten.bitwise_left_shift.Tensor	F	F	F
aten.bitwise_not	F	F	F
aten.bitwise_or.Tensor	F	F	F
aten.bitwise_right_shift.Tensor	F	F	F
aten.bitwise_xor.Tensor	F	F	F
aten.ceil	C	C	C
aten.clamp	C	C	C
aten.clamp.Tensor	C	C	C
aten.clamp_max	C	C	C
aten.clamp_max.Tensor	R	C	C
aten.clamp_min	C	C	C
aten.clamp_min.Tensor	C	C	C
aten.copysign.Tensor	R	R	R
aten.div.Scalar	C	C	C
aten.div.Scalar_mode	C	R	C
aten.div.Tensor	C	C	C
aten.div.Tensor_mode	C	C	C
aten.eq.Scalar	C	C	C
aten.eq.Tensor	C	C	C
aten.fix	C	R	R
aten.floor	C	C	C
aten.floor_divide	C	C	C
aten.floor_divide.Scalar	C	R	C
aten.fmax	R	R	R
aten.fmin	R	R	R
aten.fmod.Scalar	C	C	R

Arithmetic, comparison & logic — page 2/3

Canonical Aten compilation evidence

Operator	Graphcore	CS-3	Groq
aten.fmod.Tensor	C	C	R
aten.frac	C	R	R
aten.ge.Scalar	C	C	C
aten.ge.Tensor	C	C	C
aten.gt.Scalar	C	C	C
aten.gt.Tensor	C	C	C
aten.heaviside	R	R	R
aten.isclose	C	C	R
aten.isfinite	C	C	R
aten.isinf	C	C	R
aten.isnan	C	C	C
aten.isneginf	R	R	R
aten.isposinf	R	R	R
aten.le.Scalar	C	C	C
aten.le.Tensor	C	C	C
aten.lerp.Scalar	R	R	C
aten.lerp.Tensor	R	R	C
aten.logical_and	R	C	C
aten.logical_not	R	C	C
aten.logical_or	R	C	C
aten.logical_xor	R	C	C
aten.lt.Scalar	C	C	C
aten.lt.Tensor	C	C	C
aten.max.other	C	C	C
aten.maximum	C	C	C
aten.min.other	C	C	C
aten.minimum	C	C	C
aten.mul.Scalar	C	C	C
aten.mul.Tensor	C	C	C
aten.nan_to_num	R	R	R
aten.ne.Scalar	C	C	C
aten.ne.Tensor	C	C	C
aten.neg	C	C	C
aten.remainder.Scalar	C	C	C

Arithmetic, comparison & logic — page 3/3

Canonical ATen compilation evidence

Operator	Graphcore	CS-3	Groq
aten remainder.Tensor	C	C	C
aten round	C	C	C
aten round.decimals	C	R	C
aten rsub.Scalar	C	C	C
aten sgn	R	C	R
aten sign	C	C	C
aten signbit	R	R	R
aten sub.Scalar	C	C	C
aten sub.Tensor	C	C	C
aten trunc	C	R	R

C

 C compiled

R

 R compiler rejected

F

 F needs fixture

— not applicable

Transcendental math — page 1/2

Canonical Aten compilation evidence

Operator	Graphcore	CS-3	Groq
aten.acos	C	C	R
aten.acosh	C	C	R
aten.asin	C	C	R
aten.asinh	C	R	R
aten.atan	C	C	R
aten.atan2	C	C	R
aten.atanh	C	R	R
aten.cos	C	C	C
aten.cosh	C	C	R
aten.deg2rad	C	C	R
aten.erf	C	C	C
aten.erfinv	R	R	R
aten.exp	C	C	C
aten.exp2	R	R	R
aten.expm1	C	C	R
aten.float_power.Tensor_Tensor	C	R	R
aten.ldexp.Tensor	C	C	R
aten.log	C	C	C
aten.log10	C	C	C
aten.log1p	C	C	C
aten.log2	C	C	C
aten.logaddexp	R	R	R
aten.logaddexp2	R	R	R
aten.logit	R	R	C
aten.pow.Scalar	C	C	C
aten.pow.Tensor_Scalar	C	C	C
aten.pow.Tensor_Tensor	C	C	C
aten.rad2deg	C	C	R
aten.reciprocal	C	C	C
aten.rsqrt	C	C	C
aten.sin	C	C	C
aten.sinh	C	C	R
aten.special_expm1	C	C	R
aten.sqrt	C	C	C

Transcendental math — page 2/2

Canonical ATen compilation evidence

Operator **Graphcore** **CS-3** **Groq**

aten.square



aten.tan



aten.tanh



aten.xlogy.Tensor



 C compiled

 R compiler rejected

 F needs fixture

 — not applicable

Linear algebra

Canonical Aten compilation evidence

Operator	Graphcore	CS-3	Groq
aten.addbmm	F	F	F
aten.addmm	C	C	C
aten.baddbmm	C	C	C
aten.bilinear	F	F	F
aten.bmm	C	C	C
aten.cosine_similarity	C	C	C
aten.diag	R	R	R
aten.diag_embed	R	R	R
aten.diagonal	R	R	R
aten.diagonal_copy	R	R	R
aten.dot	C	C	R
aten.einsum	F	F	F
aten.frobenius_norm.dim	C	C	C
aten.linalg_cross	C	R	C
aten.linalg_det	R	R	R
aten.linalg_norm	C	R	R
aten.linalg_qr	R	R	R
aten.linalg_slogdet	F	F	F
aten.linalg_vector_norm	C	R	R
aten.linear	C	C	C
aten.matmul	C	C	C
aten.mm	C	C	C
aten.mv	C	R	R
aten.norm.Scalar	C	R	R
aten.norm.ScalarOpt_dim	C	R	C
aten.outer	C	C	C
aten.scaled_dot_product_attention	F	F	F
aten.t	C	C	C
aten.t_copy	C	C	R
aten.trace	R	R	R
aten.tril	R	C	C
aten.triu	R	C	C

Complex, FFT & signal

Canonical ATen compilation evidence

Operator	Graphcore	CS-3	Groq
aten.complex	R	R	R
aten.fft_fft	F	F	F
aten.fft_ifft	F	F	F
aten.fft_rfft	F	F	F
aten.imag	F	F	F
aten.polar	R	R	R
aten.real	C	C	R
aten.stft	F	F	F
aten.stft.center	F	F	F
aten.view_as_complex	F	F	F
aten.view_as_real	F	F	F

Convolution

Canonical ATen compilation evidence

Operator	Graphcore	CS-3	Groq
aten.col2im	F	F	F
aten.conv1d	C	R	C
aten.conv1d.padding	C	R	C
aten.conv2d	R	R	R
aten.conv2d.padding	C	R	C
aten.conv3d	R	R	R
aten.conv3d.padding	C	R	C
aten.conv_tbc	C	R	R
aten.conv_transpose1d	C	R	C
aten.conv_transpose2d.input	C	R	R
aten.conv_transpose3d.input	C	R	R
aten.convolution	R	R	R
aten.im2col	F	F	F

Pooling & resampling

Canonical ATen compilation evidence

Operator	Graphcore	CS-3	Groq
aten.adaptive_avg_pool1d	C	C	C
aten.adaptive_avg_pool2d	C	C	C
aten.adaptive_avg_pool3d	R	C	R
aten.adaptive_max_pool1d	R	R	R
aten.adaptive_max_pool2d	R	R	R
aten.adaptive_max_pool3d	R	R	R
aten.avg_pool1d	C	C	C
aten.avg_pool2d	R	C	C
aten.avg_pool3d	R	C	R
aten.grid_sampler	F	F	F
aten.max_pool1d	C	C	C
aten.max_pool1d_with_indices	R	R	R
aten.max_pool2d	R	C	R
aten.max_pool2d_with_indices	R	R	R
aten.max_pool3d	R	C	R
aten.max_pool3d_with_indices	R	R	R
aten.max_unpool2d	F	F	F
aten.max_unpool3d	F	F	F
aten.upsample_bilinear2d	R	R	R
aten.upsample_bilinear2d.vec	F	F	F
aten.upsample_nearest1d	C	R	R
aten.upsample_nearest1d.vec	F	F	F
aten.upsample_nearest2d	C	R	R
aten.upsample_nearest2d.vec	F	F	F

 C compiled  R compiler rejected  F needs fixture  — not applicable

Activations

Canonical ATen compilation evidence

Operator	Graphcore	CS-3	Groq
aten.celu	C	R	R
aten.dropout	C	C	C
aten.elu	C	C	C
aten.gelu	C	C	C
aten.glu	C	C	C
aten.hardshrink	C	R	C
aten.hardsigmoid	C	C	C
aten.hardswish	C	C	C
aten.hardtanh	C	C	C
aten.leaky_relu	C	C	C
aten.log_sigmoid	C	C	C
aten.log_sigmoid_forward	R	C	R
aten.log_softmax.int	C	C	C
aten.mish	C	C	C
aten.native_dropout	R	C	R
aten.prelu	F	F	F
aten.relu	C	C	C
aten.relu6	C	C	C
aten.rrelu	C	R	C
aten.rrelu_with_noise	C	R	R
aten.rrelu_with_noise_functional	F	F	F
aten.selu	C	R	C
aten.sigmoid	C	C	C
aten.silu	C	C	C
aten.softmax.int	C	C	C
aten.softplus	C	C	C
aten.softshrink	C	R	C
aten.threshold	C	C	R

Normalization

Canonical ATen compilation evidence

Operator	Graphcore	CS-3	Groq
aten.batch_norm	R	R	C
aten.group_norm	R	R	C
aten.instance_norm	R	R	C
aten.layer_norm	R	R	C
aten.native_batch_norm	R	R	R
aten.native_group_norm	F	F	F
aten.native_layer_norm	R	R	C
aten.normal_functional	C	R	R
aten.renorm	R	R	R
aten.rms_norm	F	F	F

C

 C compiled

R

 R compiler rejected

F

 F needs fixture

— not applicable

Reductions & statistics — page 1/2

Canonical Aten compilation evidence

Operator	Graphcore	CS-3	Groq
aten.all	C	C	R
aten.all.dim	C	C	R
aten.amax	C	C	C
aten.amin	C	C	C
aten.aminmax	R	R	C
aten.any	C	C	C
aten.any.dim	C	C	C
aten.any.dims	F	F	F
aten.argmax	C	C	C
aten.argmin	C	C	C
aten.argsort	C	C	C
aten.bincount	⚡	⚡	⚡
aten.count_nonzero	C	R	R
aten.count_nonzero.dim_IntList	C	R	R
aten.cumprod	C	C	R
aten.cumsum	C	C	C
aten.kthvalue	R	R	R
aten.logcumsumexp	R	R	R
aten.logsumexp	C	C	C
aten.max	C	C	C
aten.max.dim	C	C	C
aten.mean	C	C	C
aten.mean.dim	C	C	C
aten.min	C	C	C
aten.min.dim	C	C	C
aten.prod	C	R	C
aten.prod.dim_int	C	C	C
aten.sort	C	C	C
aten.std	C	C	C
aten.std.correction	R	C	C
aten.std.dim	C	C	C
aten.sum	C	C	C
aten.sum.dim_IntList	C	C	C
aten.topk	C	C	C

Reductions & statistics — page 2/2

Canonical ATen compilation evidence

Operator	Graphcore	CS-3	Groq
aten.unique_consecutive	R	R	R
aten.unique_dim	R	R	R
aten.var	C	C	C
aten.var.correction	R	C	C
aten.var.dim	C	C	C
aten.var_mean	C	R	C
aten.var_mean.correction	R	R	C
aten.var_mean.dim	C	R	C

C compiled

R compiler rejected

F needs fixture

 — not applicable

Indexing, gather & scatter — page 1/2

Canonical Aten compilation evidence

Operator	Graphcore	CS-3	Groq
aten.bucketize.Tensor	F	F	F
aten.diagonal_scatter	R	R	R
aten.embedding	F	F	F
aten.embedding_bag.padding_idx	F	F	F
aten.gather	C	C	C
aten.index.Tensor	F	F	F
aten.index.Tensor_hacked_twin	F	F	F
aten.index_put	F	F	F
aten.index_put.hacked_twin	F	F	F
aten.index_select	C	R	C
aten.masked_fill.Scalar	C	C	C
aten.masked_fill.Tensor	F	F	F
aten.masked_scatter	R	R	R
aten.masked_select	R	R	R
aten.narrow	C	C	C
aten.narrow.Tensor	F	F	F
aten.nonzero	R	R	R
aten.nonzero_numpy	R	R	R
aten.nonzero_static	F	F	F
aten.one_hot			
aten.scatter.reduce	F	F	F
aten.scatter.src	F	F	F
aten.scatter.value	F	F	F
aten.scatter_add	F	F	F
aten.scatter_reduce.two	F	F	F
aten.select.int	C	C	C
aten.select_copy.int	C	C	R
aten.select_scatter	F	F	F
aten.slice.Tensor	R	C	C
aten.slice_copy.Tensor	R	C	R
aten.slice_scatter	R	C	R
aten.where.Scalar			
aten.where.Scalar0ther	C	C	R
aten.where.ScalarSelf	C	C	R

Indexing, gather & scatter — page 2/2

Canonical ATen compilation evidence

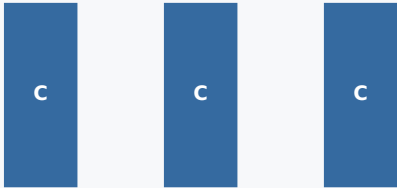
Operator

Graphcore

CS-3

Groq

aten.where.self



Operator	Graphcore	CS-3	Groq
aten.alias	R	C	C
aten.alias_copy	R	C	R
aten.as_strided	R	R	C
aten.as_strided_copy	R	R	R
aten.as_strided_scatter	F	F	F
aten.atleast_1d	C	C	C
aten.atleast_2d	C	C	C
aten.broadcast_tensors	C	C	C
aten.broadcast_to	C	C	C
aten.cat	C	C	C
aten.channel_shuffle	R	R	R
aten.chunk	C	C	C
aten.clone	C	C	C
aten.column_stack	C	C	R
aten.constant_pad_nd	F	F	F
aten.contiguous	C	C	C
aten.copy	R	C	R
aten.cpu	R	R	C
aten.cuda	F	F	F
aten.detach	C	C	C
aten.detach_copy	C	C	R
aten.empty_like	C	C	R
aten.expand	C	C	C
aten.expand_as	C	C	C
aten.expand_copy	C	C	R
aten.fill.Scalar	C	R	R
aten.fill.Tensor	F	F	F
aten.flatten.using_ints	C	C	C
aten.flip	C	R	C
aten.fliplr	C	R	R
aten.flipud	C	R	R
aten.full_like	C	C	R
aten.hstack	C	C	R
aten.lift_fresh_copy	C	C	R

Operator	Graphcore	CS-3	Groq
aten.meshgrid	F	F	F
aten.meshgrid.indexing	F	F	F
aten.movedim.int	C	C	C
aten.new_empty	C	C	R
aten.new_empty_strided	F	F	F
aten.new_full	C	C	R
aten.new_ones	C	C	R
aten.new_zeros	C	C	R
aten.numpy_T	C	C	C
aten.ones_like	C	C	R
aten.pad	F	F	F
aten.permute	C	C	C
aten.permute_copy	C	C	R
aten.pixel_shuffle	C	R	R
aten.pixel_unshuffle	C	C	C
aten.reflection_pad1d	C	R	R
aten.reflection_pad2d	F	F	F
aten.reflection_pad3d	F	F	F
aten.repeat	C	C	C
aten.repeat_interleave.Tensor	F	F	F
aten.repeat_interleave.self_int	F	F	F
aten.replication_pad1d	C	R	R
aten.replication_pad2d	F	F	F
aten.replication_pad3d	F	F	F
aten.reshape	C	C	C
aten.reshape_as	C	C	C
aten.resize	R	R	R
aten.roll	F	F	F
aten.rot90	C	R	R
aten.split.Tensor	C	C	C
aten.split.sizes	C	C	C
aten.split_copy.Tensor	C	C	R
aten.split_with_sizes	C	C	C
aten.split_with_sizes_copy	C	C	R

Shape, layout & construction — page 3/3

Canonical Aten compilation evidence

Operator	Graphcore	CS-3	Groq
aten.squeeze	C	C	C
aten.squeeze.dim	C	C	C
aten.squeeze_copy	C	C	R
aten.squeeze_copy.dim	C	C	R
aten.stack	C	C	C
aten.tensor_split.sections	C	C	R
aten.tile	R	R	R
aten.to.device	R	R	C
aten.to.dtype	C	C	C
aten.to.dtype_layout	R	R	C
aten.to.other	C	C	C
aten.to.prim_Device	R	R	C
aten.transpose.int	C	C	C
aten.transpose_copy.int	C	C	R
aten.type_as	C	C	C
aten.unbind.int	C	C	C
aten.unbind_copy.int	C	R	R
aten.unflatten.int	F	F	F
aten.unfold	F	F	F
aten.unfold_copy	F	F	F
aten.unsqueeze	C	C	C
aten.unsqueeze_copy	C	C	R
aten.view	C	C	C
aten.view.dtype	F	F	F
aten.view_copy	C	C	R
aten.view_copy.dtype	F	F	F
aten.zero	R	R	R
aten.zeros_like	C	C	R

Loss functions

Canonical ATen compilation evidence

Operator	Graphcore	CS-3	Groq
aten.binary_cross_entropy	C	R	R
aten.binary_cross_entropy_with_logits	C	C	C
aten.cosine_embedding_loss	C	C	R
aten.cross_entropy_loss	R	R	R
aten.kl_div	C	C	C
aten.l1_loss	C	C	R
aten.mse_loss	C	C	C
aten.nll_loss2d_forward	F	F	F
aten.nll_loss_forward	F	F	F
aten.poisson_nll_loss	C	C	R

C

 C compiled

R

 R compiler rejected

F

 F needs fixture

— not applicable

Random & sampling

Canonical ATen compilation evidence

Operator	Graphcore	CS-3	Groq
aten.bernoulli	F	F	F
aten.bernoulli.Tensor	F	F	F
aten.bernoulli.p	C	R	R
aten.exponential	C	R	R
aten.multinomial	F	F	F
aten.rand_like	C	C	R
aten.randn_like	C	C	R
aten.random	R	R	R
aten.random.from	R	R	R
aten.uniform	C	R	R

C

 C compiled

R

 R compiler rejected

F

 F needs fixture

— not applicable

Quantization

Canonical ATen compilation evidence

Operator	Graphcore	CS-3	Groq
aten.dequantize.self	R	R	R
aten.dequantize.tensor	R	R	R
aten.fake_quantize_per_channel_affine	R	R	R
aten.fake_quantize_per_channel_affine_cachemask	R	R	R
aten.fake_quantize_per_tensor_affine	R	R	R
aten.fake_quantize_per_tensor_affine.tensor_qparams	F	F	F
aten.fake_quantize_per_tensor_affine_cachemask	R	R	R
aten.int_repr	—	—	—
aten.quantize_per_channel	F	F	F
aten.quantize_per_tensor	F	F	F

 C compiled

 R compiler rejected

 F needs fixture

 — not applicable